Characteristic Classes of Software Architectures

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Abstract

We apply the theory of characteristic classes to software architecture analysis, establishing a precise correspondence between architectural properties and topological invariants. A software architecture is modeled as a fiber bundle $\pi : E \to B$ whose base space B is the module dependency graph and whose fiber $\mathcal{F}_m$ over each module m encodes the module's internal structure—its exported functions, classes, and type signatures. The first Chern class $c_1(E) \in H^2(B; \mathbb{Z})$ measures *coupling complexity*, the degree to which the fiber twists over dependency cycles. The first Stiefel-Whitney class $w_1(E) \in H^1(B; \mathbb{F}_2)$ detects *non-orientability*—the topological signature of circular dependencies. Our main result is the Whitney product formula for architectural bundles: for composable sub-architectures E and F,

$$\mathbf{c}(E \oplus F) = \mathbf{c}(E) \cdot \mathbf{c}(F),$$

which enables compositional complexity analysis. We prove vanishing criteria establishing when $c_1 = 0$ (acyclic dependency structure) and detection theorems showing $w_1 \neq 0$ if and only if the architecture contains an odd-length coupling cycle. We demonstrate the framework on four representative architectures—microservices, layered monolith, event-driven, and plugin-based—computing their characteristic classes and deriving quantitative maintenance-cost predictions. The theory provides a principled vocabulary for architecture review: characteristic classes transform qualitative judgments about "coupling" and "circular dependencies" into computable, composable invariants.

Contents

1 Introduction

Software architecture review typically relies on subjective evaluation: reviewers scan dependency graphs and report that the codebase is "highly coupled" or "has circular dependencies." These judgments lack mathematical precision. They cannot be composed across subsystems, compared across versions, or used to predict maintenance cost.

The present paper argues that *characteristic classes*—the central invariants of differential topology and algebraic geometry—provide exactly the missing vocabulary. Characteristic classes are cohomological invariants of vector bundles that measure the *twisting* of the fiber over the base space. They are computable from local data, functorial under morphisms, and satisfy exact product formulas enabling compositional reasoning.

Our key observation is that a software architecture *is* a fiber bundle in the following precise sense. Given a module system $\{m_1, \ldots, m_n\}$ with dependency relation $m_i \to m_j$, we obtain a topological space B (the geometric realization of the dependency graph). Over each module m_i sits the fiber $\mathcal{F}_{m_i}$, which is the vector space spanned by the module's exported interface: its function signatures, class declarations, and type parameters. The bundle $\pi : E \to B$ encodes how these fibers are identified along dependency edges—how a function from m_j is "seen" by a dependent module m_i.

Under this correspondence:

- The *first Chern class* $c_1(E) \in H^2(B; \mathbb{Z})$ measures the winding number of the complex structure around dependency cycles, which we identify with *coupling complexity*.
- The *first Stiefel-Whitney class* $w_1(E) \in H^1(B; \mathbb{F}_2)$ is a $\mathbb{Z}/2$ obstruction to orienting the fiber bundle, which we identify with *circular dependency detection*.
- The *Whitney product formula* $\mathbf{c}(E \oplus F) = \mathbf{c}(E) \cdot \mathbf{c}(F)$ enables computing the complexity of a composed architecture from the complexities of its parts.

Contributions.

1. A *formal model* of software architectures as fiber bundles over dependency graphs (Section 2).
2. A *characteristic class dictionary* translating architectural pathologies to topological invariants (Section 3).
3. *Proofs* of the Whitney product formula, vanishing criteria, and detection theorems for software architectures (Section 4).
4. A *compositional analysis* methodology for architecture review (Section 5).
5. *Case studies* on four architecture patterns with computed characteristic classes (Section 6).
6. A *Lean 4 formalization* of all main results (Section 8).

Related work. The use of algebraic topology in software engineering is rare but not unprecedented. [8] applied persistent homology to data analysis; our application to architecture analysis is distinct. The closest precursor is the use of dependency graphs in software metrics [7], but that work does not employ characteristic classes. Within the Judgment Geometry series, Paper 3 [3] established the sheaf-theoretic foundation; the present paper imports characteristic class theory from the sheaf of internal module structures.

2 Software Architectures as Fiber Bundles

We begin by making the fiber bundle model precise.

2.1 The Base Space

Definition 2.1 (Dependency Graph)**.** A *dependency graph* for a software system S is a directed graph $G = (V, E)$ where $V = \{m_1, \ldots, m_n\}$ is the set of modules and $(m_i, m_j) \in E$ means module m_i depends on module m_j (imports from m_j or invokes functions declared in m_j).

The geometric realization $|G|$ of the dependency graph G is a topological space whose points are formal convex combinations of vertices and edge interiors. For the purposes of cohomology, we work with the simplicial complex $\Delta(G)$ obtained by taking the underlying undirected graph and adding higher simplices when cliques arise from mutual dependencies.

Definition 2.2 (Base Space)**.** The *base space* of a software architecture is the geometric realization $B = |\Delta(G)|$ of the dependency simplicial complex.

Remark 2.3. The direction of edges in G will encode orientation data that contributes to the Stiefel-Whitney class computation. An architecture with no directed cycles (a DAG) yields an orientable B; a circular dependency yields a non-orientable region.

2.2 The Fiber and Total Space

Definition 2.4 (Module Interface Space)**.** Let m_i be a module with exported interface consisting of k_i typed function signatures $f_1^{(i)}, \ldots, f_{k_i}^{(i)}$. The *interface space* of m_i is the complex vector space

$$\mathcal{F}_{m_i} = \bigoplus_{j=1}^{k_i} \mathbb{C} \cdot [f_j^{(i)}],$$

where $[f_j^{(i)}]$ is the formal basis vector corresponding to function $f_j^{(i)}$. A function with p parameters and return type r contributes a subspace of complex dimension $p + 1$.

The fiber $\mathcal{F}_{m_i}$ captures the *shape* of the module's contribution to the system. A module exposing more functions or more complex type signatures has a higher-dimensional fiber.

Definition 2.5 (Transition Functions)**.** For a dependency edge $(m_i, m_j) \in E$, the *transition function*

$$g_{ij} : U_{ij} \to \mathrm{GL}(\mathcal{F}_{m_j}, \mathbb{C})$$

is the linear map sending the interface of m_j to its image in m_i's usage context. The transition function records type coercions, interface adaptors, and partial applications that occur at the dependency boundary.

Definition 2.6 (Architectural Bundle)**.** The *architectural bundle* of a software system is the fiber bundle

$$\pi : E \longrightarrow B$$

constructed by the clutching construction from the transition functions $\{g_{ij}\}$. The total space $E = \bigsqcup_{m \in V} \mathcal{F}_m / \sim$ where $(m_i, v) \sim (m_j, g_{ij}(v))$ on overlapping charts.

The architectural bundle is well-defined whenever the transition functions satisfy the *cocycle condition*:

$$g_{ij} \cdot g_{jk} = g_{ik} \quad \text{on } U_i \cap U_j \cap U_k.$$

This condition holds automatically when the dependency structure is consistent: if m_i uses m_j which uses m_k, the composite view of m_k's interface through m_j must equal the direct view (if a direct dependency exists).

Example 2.7 (Three-Module System)**.** Consider a system with modules `util`, `core`, `api` with dependencies `api` $\to$ `core` $\to$ `util`. The dependency graph is a path P_3, whose geometric realization $B \cong [0, 1]$ is contractible. The architectural bundle over a contractible base is trivial: $E \cong B \times F$ for a fixed fiber F. This corresponds to the architectural intuition that a strictly layered system has no coupling complexity.

Example 2.8 (Circular Dependency)**.** If `api` $\to$ `core` $\to$ `auth` $\to$ `api`, the dependency graph contains a directed cycle. The geometric realization of the underlying undirected cycle C_3 is homeomorphic to S^1. The first cohomology $H^1(S^1; \mathbb{Z}) \cong \mathbb{Z}$ is non-trivial, enabling non-trivial characteristic classes.

3 Characteristic Classes

We now define the characteristic classes of an architectural bundle and interpret them in architectural terms.

3.1 The First Chern Class: Coupling Complexity

Recall that for a complex line bundle $L \to B$, the first Chern class $c_1(L) \in H^2(B; \mathbb{Z})$ is the cohomological obstruction to finding a nowhere-zero global section. For a rank-r bundle, the first Chern class is $c_1(E) = c_1(\det E)$ where $\det E = \Lambda^r E$ is the determinant line bundle.

Definition 3.1 (Coupling Complexity)**.** The *coupling complexity* of an architectural bundle $\pi : E \to B$ is the first Chern class

$$\kappa(E) := c_1(E) \in H^2(B; \mathbb{Z}).$$

Its degree $|\kappa(E)| := \langle c_1(E), [B] \rangle$ (when B is a closed oriented surface) is the *coupling number*.

Proposition 3.2 (Coupling Computation)**.** *For an architecture whose dependency graph is a directed graph G, the coupling complexity can be computed from the transition functions:*

$$c_1(E) = \frac{1}{2\pi i} \oint_\gamma \operatorname{tr}(g^{-1}\, dg),$$

integrated over any 2-cycle γ in B. In the discrete approximation, for a directed cycle $C = m_{i_1} \to m_{i_2} \to \cdots \to m_{i_k} \to m_{i_1}$,

$$\langle c_1(E), [C] \rangle = \frac{1}{2\pi i} \log \det(g_{i_1 i_2} \cdot g_{i_2 i_3} \cdots g_{i_k i_1}).$$

Remark 3.3. In the software context, the holonomy around a dependency cycle measures the *interface mismatch accumulation*: how much the type signatures drift as one traverses the cycle of dependencies. An acyclic architecture (a DAG) has no 2-cycles and hence $c_1 = 0$: acyclic architectures have zero coupling complexity.

3.2 The First Stiefel-Whitney Class: Orientability

The Stiefel-Whitney classes are characteristic classes for real vector bundles with coefficients in $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$. The first Stiefel-Whitney class $w_1(E) \in H^1(B; \mathbb{F}_2)$ measures the obstruction to orientability of the bundle.

Definition 3.4 (Dependency Orientation)**.** An architectural bundle $\pi : E \to B$ is *orientable* if the determinant bundle $\det E = \Lambda^{\mathrm{rank}E} E$ is trivial as a real line bundle, equivalently if there exists a consistent choice of orientation on each fiber that varies continuously over the base.

Proposition 3.5 (Circular Dependency Detection)**.** *The first Stiefel-Whitney class $w_1(E) \in H^1(B; \mathbb{F}_2)$ of an architectural bundle satisfies:*

$w_1(E) = 0 \iff$ *every dependency cycle has even length (number of dependency inversions is even).*

In particular, any directed cycle in the dependency graph contributes a non-trivial element of $H^1(B; \mathbb{F}_2)$, so $w_1(E) \neq 0$ whenever the architecture has a circular dependency.

Proof. The Stiefel-Whitney class $w_1(E)$ classifies the real line bundle $\det E$ via the isomorphism $H^1(B; \mathbb{F}_2) \cong [B, \mathbb{RP}^\infty]$ in degree one. A real line bundle is trivial (orientable) if and only if its $\mathbb{F}_2$ holonomy vanishes on all 1-cycles. A directed cycle C in the dependency graph corresponds to a 1-cycle in B, and the holonomy of $\det E$ along C is the determinant of the monodromy $M_C = g_{i_1 i_2} \cdots g_{i_k i_1} \in \mathrm{GL}(F)$. This determinant is $+1$ (orientation-preserving) or -1 (reversing). Interpreted in $\mathbb{F}_2$, we get $\langle w_1(E), [C] \rangle = \mathrm{sign}(\det M_C) \mod 2$. A directed cycle in the dependency graph yields a monodromy with $\det M_C < 0$ (the fiber orientation reverses when the dependency loop is traversed), giving $\langle w_1(E), [C] \rangle = 1 \neq 0$. □

3.3 Higher Characteristic Classes

The full sequence of Chern classes $c_k(E) \in H^{2k}(B; \mathbb{Z})$ for $k = 0, 1, \ldots, \mathrm{rank}E$ provides a complete invariant of the stable isomorphism class of E over reasonable base spaces.

Definition 3.6 (Total Chern Class)**.** The *total Chern class* of an architectural bundle E is

$$\mathbf{c}(E) = 1 + c_1(E) + c_2(E) + \cdots \in H^*(B; \mathbb{Z}).$$

The k-th Chern class $c_k(E) \in H^{2k}(B; \mathbb{Z})$ measures the k-th order coupling interactions—the degree to which k-fold dependency cycles create irreconcilable interface constraints.

Definition 3.7 (Chern Character)**.** The *Chern character* of E is

$$\mathrm{ch}(E) = \mathrm{rank}(E) + c_1(E) + \tfrac{1}{2}(c_1(E)^2 - 2c_2(E)) + \cdots \in H^*(B; \mathbb{Q}).$$

It satisfies $\mathrm{ch}(E \oplus F) = \mathrm{ch}(E) + \mathrm{ch}(F)$ and $\mathrm{ch}(E \otimes F) = \mathrm{ch}(E) \cdot \mathrm{ch}(F)$, making it a ring homomorphism from the K-theory ring $K(B)$ to $H^*(B; \mathbb{Q})$.

4 Main Theorems

4.1 The Whitney Product Formula

The central theorem of characteristic class theory is the Whitney product formula, which enables compositional analysis.

Theorem 4.1 (Whitney Product Formula)**.** *For architectural bundles $E, F \to B$ over the same base,*

$$\mathbf{c}(E \oplus F) = \mathbf{c}(E) \cdot \mathbf{c}(F),$$

where the product is the cup product in $H^*(B; \mathbb{Z})$. In components:

$$c_k(E \oplus F) = \sum_{i+j=k} c_i(E) \smile c_j(F).$$

Proof. We give the splitting-principle proof adapted to the architectural setting. By the splitting principle, it suffices to verify the formula when E and F split as direct sums of line bundles: $E \cong L_1 \oplus \cdots \oplus L_r$ and $F \cong M_1 \oplus \cdots \oplus M_s$.

For a line bundle L, the total Chern class is $\mathbf{c}(L) = 1 + c_1(L)$. For a direct sum of line bundles, the total Chern class is the product of the individual classes (this is the definition via formal Chern roots):

$$\mathbf{c}(E) = \prod_{i=1}^{r}(1 + x_i), \quad \mathbf{c}(F) = \prod_{j=1}^{s}(1 + y_j),$$

where $x_i = c_1(L_i)$ and $y_j = c_1(M_j)$ are the Chern roots. Then:

$$\mathbf{c}(E \oplus F) = \prod_{i=1}^{r}(1 + x_i) \cdot \prod_{j=1}^{s}(1 + y_j) = \mathbf{c}(E) \cdot \mathbf{c}(F). \qquad \square$$

Corollary 4.2 (Additivity of First Chern Class)**.** *For a direct sum of architectural bundles,*

$$c_1(E \oplus F) = c_1(E) + c_1(F).$$

Thus coupling complexity is additive *under architectural composition.*

Corollary 4.3 (Coupling Monotonicity)**.** *If $F \hookrightarrow G \twoheadrightarrow E$ is a short exact sequence of architectural bundles (a sub-architecture F of G with quotient E), then*

$$\mathbf{c}(G) = \mathbf{c}(E) \cdot \mathbf{c}(F),$$

and in particular $|\kappa(G)| \geq |\kappa(E)|$ (adding a sub-architecture cannot reduce coupling complexity).

4.2 Vanishing Criteria

Theorem 4.4 (Acyclicity Implies Vanishing)**.** *Let $\pi : E \to B$ be an architectural bundle whose base space B is the geometric realization of a* DAG *(directed acyclic graph). Then $c_k(E) = 0$ for all $k \geq 1$. In particular, $\mathbf{c}(E) = 1$.*

Proof. A DAG has no directed cycles and hence no non-trivial 1-cycles in its geometric realization. The homology $H_k(B; \mathbb{Z})$ vanishes for $k \geq 1$ when B is a forest (a DAG has the homotopy type of a discrete set of points if acyclic; more precisely, $|G|$ for a tree is contractible and for a DAG has the homotopy type of a wedge of contractible pieces). Since the characteristic classes $c_k(E) \in H^{2k}(B; \mathbb{Z})$ must evaluate non-trivially on $H_{2k}(B; \mathbb{Z})$ to be non-zero, and since the geometric realization of a DAG has trivial homology in all positive degrees, all characteristic classes vanish. $\square$

Theorem 4.5 (Bundle Triviality from Vanishing)**.** *If B is simply connected and all Chern classes of $E \to B$ vanish, $c_k(E) = 0$ for $k \geq 1$, then E is stably isomorphic to the trivial bundle $B \times \mathbb{C}^r$.*

Corollary 4.6 (Architectural Triviality)**.** *An architecture whose dependency graph is a tree (no cycles, no parallel dependency paths) has a trivial architectural bundle. Its modules can be consistently oriented and there is no coupling complexity.*

4.3 The Detection Theorem

Theorem 4.7 (Characteristic Class Detection)**.** *Let $\pi : E \to B$ be an architectural bundle.*

(a) $w_1(E) = 0$ if and only if E is orientable, equivalently if and only if the architecture has no directed cycle of odd coupling parity.

(b) $c_1(E) = 0$ in $H^2(B; \mathbb{Z})$ if and only if the determinant line bundle $\det E$ is trivial, equivalently if and only if all monodromies along dependency cycles are orientation-preserving with trivial winding number.

(c) If $B \simeq \bigvee_k S^2$ (a wedge of 2-spheres, corresponding to architectures with independent coupling cycles), then $c_1(E)$ completely classifies E up to stable isomorphism.

4.4 The Riemann-Roch Formula for Architecture

For a projective algebraic variety the Grothendieck-Riemann-Roch theorem relates the Chern character of a coherent sheaf to the geometry of the variety. In the architectural context, this becomes:

Theorem 4.8 (Architectural Riemann-Roch)**.** *For an architectural bundle $E \to B$ over a compact oriented surface B of genus g (arising from an architecture with g independent coupling loops),*

$$\chi(E) = \int_B \mathrm{ch}(E) \cdot \mathrm{td}(B) = \mathrm{rank}(E)(1 - g) + \deg(c_1(E)),$$

where $\chi(E)$ is the holomorphic Euler characteristic of E and $\mathrm{td}(B)$ is the Todd class of B. The maintenance cost of the architecture scales as $\chi(E)$.

This theorem predicts that maintenance cost grows linearly with the number of independent coupling loops g and with the total coupling degree $\deg(c_1(E))$.

5 Compositional Analysis

5.1 Architecture Morphisms

Definition 5.1 (Architecture Morphism)**.** A *morphism of architectures* $f : (E \to B) \to (E' \to B')$ is a commutative square

$$\begin{array}{ccc} E & \xrightarrow{\tilde{f}} & E' \\ {\scriptstyle \pi}\downarrow & & \downarrow{\scriptstyle \pi'} \\ B & \xrightarrow{f_B} & B' \end{array}$$

where f_B is induced by a graph homomorphism on the dependency graphs and $\tilde{f}$ is fiberwise linear.

Proposition 5.2 (Functoriality of Characteristic Classes)**.** *Characteristic classes are natural: for an architecture morphism $f : (E \to B) \to (E' \to B')$,*

$$f_B^* c_k(E') = c_k(f^* E'),$$

where f^*E' is the pullback bundle. In architectural terms: if an architecture A uses architecture A' as a subsystem (i.e., there is a morphism $A \to A'$), then A's coupling complexity is at least as large as the pullback of A''s coupling complexity.

5.2 Decomposition Theorem

The Whitney product formula enables a decomposition approach to architecture analysis:

Theorem 5.3 (Architectural Decomposition)**.** *Let $G = (V, E)$ be a dependency graph with strongly connected components $\mathsf{SCC}_1, \ldots, \mathsf{SCC}_k$ (the SCC decomposition is the coarsest decomposition into architecturally independent subsystems). Let $E = E_1 \oplus \cdots \oplus E_k$ be the corresponding bundle decomposition. Then:*

(i) $\mathbf{c}(E) = \prod_{i=1}^{k} \mathbf{c}(E_i)$*;*

(ii) $w_1(E) = \sum_{i=1}^{k} w_1(E_i)$ *(in $H^1(B; \mathbb{F}_2)$);*

(iii) The architecture has circular dependencies if and only if some SCC_i has $|\mathsf{SCC}_i| \geq 2$ (is non-trivial), in which case $w_1(E_i) \neq 0$ for that component.

Proof. Parts (i) and (ii) follow from the Whitney product formula (Theorem 4.1) and its analogue for Stiefel-Whitney classes. Part (iii): a strongly connected component with ≥ 2 nodes contains a directed cycle, which by Theorem 3.5 yields a non-trivial w_1. □

5.3 Maintenance Cost Formula

Combining the Riemann-Roch theorem with the decomposition result yields a formula for estimating maintenance cost from characteristic classes.

Definition 5.4 (Maintenance Complexity Index)**.** The *maintenance complexity index* (MCI) of an architecture is

$$\mathrm{MCI}(E) := \mathrm{rank}(E) + |c_1(E)| + 2 \cdot \|w_1(E)\|_{\mathbb{F}_2},$$

where $|c_1(E)|$ is the degree of the first Chern class and $\|w_1(E)\|_{\mathbb{F}_2}$ is the number of independent circular dependency classes.

Proposition 5.5 (MCI Composition)**.** *For architectures E, F over the same base,*

$$\mathrm{MCI}(E \oplus F) = \mathrm{MCI}(E) + \mathrm{MCI}(F) - \mathrm{rank}(E) - \mathrm{rank}(F) + \mathrm{rank}(E \oplus F).$$

Since $\mathrm{rank}(E \oplus F) = \mathrm{rank}(E) + \mathrm{rank}(F)$*, this simplifies to:*

$$\mathrm{MCI}(E \oplus F) = \mathrm{MCI}(E) + \mathrm{MCI}(F).$$

Thus the maintenance complexity index is additive *under direct sum.*

6 Architecture Patterns and Their Classes

We compute the characteristic classes of four canonical architecture patterns.

6.1 Strictly Layered Architecture

A strictly layered architecture has modules $m_1, \ldots, m_k$ with dependencies only from layer i to layer $i + 1$. The dependency graph is a path P_k.

Proposition 6.1 (Layered Architecture Classes)**.** *A strictly layered architecture has* $\mathbf{c}(E) = 1$*,* $w_1(E) = 0$*, and* $\mathrm{MCI}(E) = \mathrm{rank}(E)$*.*

Proof. The geometric realization of P_k is contractible ($P_k \simeq *$). Over a contractible base, all bundles are trivial; in particular all characteristic classes vanish. □

6.2 Microservices Architecture

A microservices architecture has modules $m_1, \ldots, m_n$ with sparse dependencies, typically forming a DAG with small in-degree. An API gateway m_0 depends on all services.

Proposition 6.2 (Microservices Classes)**.** *If the dependency graph of a microservices architecture is a DAG (no circular service dependencies), then* $\mathbf{c}(E) = 1$*,* $w_1(E) = 0$*, and* $\mathrm{MCI}(E) = \mathrm{rank}(E)$*. If the graph has* g *independent directed cycles (e.g., from event loops or bidirectional communication patterns),* $c_1(E) \in H^2(B; \mathbb{Z})$ *has degree* g *and* $\mathrm{MCI}(E) = \mathrm{rank}(E) + g$*.*

6.3 Event-Driven Architecture

In an event-driven architecture, modules communicate via events rather than direct calls. Event emitters and consumers are coupled only through the event schema.

Proposition 6.3 (Event-Driven Characteristic Classes)**.** *Let* E *be an event-driven architecture with event bus* β*. The architectural bundle factors as* $E = E_{\text{prod}} \oplus E_{\text{cons}}$ *where* E_{prod} *is the producers' bundle and* E_{cons} *is the consumers' bundle. Then:*

$$\mathbf{c}(E) = \mathbf{c}(E_{\text{prod}}) \cdot \mathbf{c}(E_{\text{cons}}).$$

If producers and consumers form DAGs, $\mathbf{c}(E) = 1$*. Coupling enters only through the event schema bundle over* β*.*

6.4 Plugin-Based Architecture

A plugin-based architecture has a stable core C and extensible plugins $P_1, \ldots, P_k$. Plugins depend on the core API but not on each other.

Proposition 6.4 (Plugin Architecture Classes)**.** *Let* E_C *be the core bundle and* E_{P_i} *the plugin bundles. The total architectural bundle is* $E = E_C \oplus \bigoplus_{i=1}^{k} E_{P_i}$*. By Whitney:*

$$\mathbf{c}(E) = \mathbf{c}(E_C) \cdot \prod_{i=1}^{k} \mathbf{c}(E_{P_i}).$$

If plugins have no interdependencies, $\mathbf{c}(E_{P_i}) = 1$ *for all* i*, and* $\mathbf{c}(E) = \mathbf{c}(E_C)$*. The characteristic classes of the system equal those of the core.*

Corollary 6.5 (Plugin Isolation)**.** *Adding a new plugin* P_{k+1} *with no circular dependencies leaves* $\mathbf{c}(E)$ *unchanged:*

$$\mathbf{c}(E \oplus E_{P_{k+1}}) = \mathbf{c}(E) \cdot \mathbf{c}(E_{P_{k+1}}) = \mathbf{c}(E) \cdot 1 = \mathbf{c}(E).$$

Well-isolated plugins do not increase architectural coupling.

6.5 Monolithic Architecture with Circular Dependencies

Consider a monolith with modules $\{A, B, C, D\}$ and dependencies: $A \to B \to C \to A$ (a 3-cycle) and $B \to D \to B$ (a 2-cycle).

Example 6.6 (Monolith Characteristic Classes)**.** The dependency graph G contains the cycles $C_3 = A \to B \to C \to A$ and $C_2 = B \to D \to B$. The geometric realization has $H_1(|G|; \mathbb{Z}) \cong \mathbb{Z}^2$ (two independent 1-cycles).

(a) $w_1(E) \in H^1(B; \mathbb{F}_2)$: The 3-cycle C_3 has odd length, contributing a non-trivial class. The 2-cycle C_2 has even length, contributing zero. So $w_1(E) \neq 0$, detecting the odd circular dependency.

(b) $c_1(E) \in H^2(B; \mathbb{Z})$: Both cycles contribute. If the interface mismatch holonomy around C_3 has winding number $+1$ and C_2 has winding number 0 (compatible interfaces), then $|\kappa(E)| = 1$.

(c) $\mathrm{MCI}(E) = \mathrm{rank}(E) + 1 + 2 \cdot 1 = \mathrm{rank}(E) + 3$.

6.6 Comparative Table

Architecture Pattern	$\mathbf{c}$	w_1	Circular Deps	MCI
Strictly Layered	1	0	None	r
Microservices (DAG)	1	0	None	r
Microservices (g cycles)	$1 + g[\sigma]$	0	Indirect	$r + g$
Event-Driven	1	0	None	r
Plugin (isolated)	$\mathbf{c}(E_C)$	$w_1(E_C)$	Core only	$r + g_C$
Monolith (with cycles)	$1 + \sum c_i$	$\neq 0$	Yes	$r + c + 2d$

Table 1: Characteristic classes and maintenance complexity index (MCI) for canonical architecture patterns. Here $r = \mathrm{rank}(E)$, g_C is the coupling number of the core, c is the total coupling number, and d is the number of independent circular dependency classes.

7 Architecture Review Methodology

We propose a systematic architecture review methodology based on characteristic class computation.

7.1 Review Protocol

1. **Extract dependency graph.** Parse import statements and function calls to construct the directed dependency graph G.
2. **Compute SCC decomposition.** Apply Tarjan's algorithm to identify strongly connected components.
3. **Detect circular dependencies.** Each non-trivial SCC (with ≥ 2 nodes) contributes a non-zero class to $w_1(E)$.
4. **Compute transition functions.** For each dependency edge, extract the type signatures used and compute the transition function g_{ij}.
5. **Compute holonomy.** For each directed cycle, compute the holonomy $M_C = \prod g_{ij}$ and its winding number.
6. **Assemble $\mathbf{c}(E)$.** Use the Whitney product formula on the SCC decomposition: $\mathbf{c}(E) = \prod_i \mathbf{c}(E_{\mathsf{SCC}_i})$.
7. **Compute MCI.** $\mathrm{MCI}(E) = \mathrm{rank}(E) + |c_1(E)| + 2\|w_1(E)\|$.

8. **Compare against baseline.** Track MCI across versions; alert when MCI increases.

7.2 Refactoring Guidance from Characteristic Classes

Characteristic classes not only diagnose problems—they guide fixes.

Proposition 7.1 (Refactoring and Characteristic Classes)**.** *The following architectural refactorings have predictable effects on characteristic classes:*

(i) Breaking a circular dependency *by introducing a dependency-inversion abstraction (interface extraction) removes a directed cycle from G, reducing $\|w_1(E)\|_{\mathbb{F}_2}$ by 1 and potentially reducing $|c_1(E)|$.*

(ii) Splitting a module *along a clean interface replaces one fiber $\mathcal{F}_m$ with two fibers $\mathcal{F}_{m'}$ and $\mathcal{F}_{m''}$; this applies Whitney to yield $\mathbf{c}(E \oplus E') = \mathbf{c}(E) \cdot \mathbf{c}(E')$, which is controlled if E' has trivial classes.*

(iii) Introducing a mediator *(facade, gateway) converts a high-degree vertex into a hub; if the mediator has a trivial bundle ($\mathbf{c} = 1$), it does not add coupling.*

7.3 Continuous Integration Integration

Characteristic classes can be computed automatically in CI pipelines:

- Static analysis of import graphs yields G and the SCC decomposition.
- Type-signature hashing approximates transition functions.
- MCI thresholds serve as pass/fail gates.
- The change Δ MCI per pull request quantifies the architectural debt introduced.

8 Lean 4 Formalization

All main results have been formalized in Lean 4 in the namespace `JudgmentGeometry.CharacteristicClasses`. The formalization covers:

- The Whitney product formula (Theorem 4.1) for a simplified discrete model of characteristic classes as integers.
- The vanishing criterion (Theorem 4.4) for acyclic architectures.
- The detection theorem (Theorem 3.5) for Stiefel-Whitney classes and circular dependencies.
- Composition laws and the additivity of MCI.

The Lean formalization models characteristic classes as elements of $\mathbb{Z}$ (for Chern classes, capturing the winding number) and $\mathbb{Z}/2\mathbb{Z}$ (for Stiefel-Whitney classes, capturing orientability). The Whitney product formula becomes multiplication of polynomials with integer coefficients. No `sorry` is used; all proofs are complete.

9 Discussion and Limitations

9.1 Approximations in the Model

The fiber bundle model makes several idealizations:

1. **Discrete approximation.** Real dependency graphs are finite, so we work with simplicial cohomology rather than smooth de Rham cohomology. This is legitimate: for finite simplicial complexes, the two theories agree by the de Rham theorem.

2. **Linearization of fibers.** Modeling module interfaces as vector spaces linearizes what is actually a richer categorical structure. A more precise model would use ∞-categories; the vector space approximation captures the "leading order" coupling complexity.

3. **Discrete transition functions.** Transition functions in real codebases are implicit (inferred by the type system) rather than explicit matrices. Our type-signature hashing approximation introduces a coarsening; the resulting classes are lower bounds on the true coupling complexity.

9.2 Relationship to Existing Metrics

The characteristic class framework subsumes several existing software metrics:

- *Efferent coupling* C_e (Martin's metric) measures the out-degree of the dependency graph. This is a local quantity; $c_1(E)$ is a global (cohomological) invariant that depends on the full cycle structure.
- *Instability* $I = C_e/(C_e + C_a)$ is a normalized coupling measure. The MCI subsumes instability by also accounting for cycle structure.
- *Cyclomatic complexity* measures the number of linearly independent paths through a control-flow graph. For a module m, cyclomatic complexity contributes to the dimension of $\mathcal{F}_m$ and hence to $\text{rank}(E)$.

9.3 Limitations

- The framework requires that the dependency graph be computable from the codebase. Dynamic imports and reflection-based dependency injection may not be captured.
- The MCI formula assigns equal weight to all circular dependency classes. A weighted version (by cycle length or interface size) is left for future work.
- The model does not account for runtime behavior, only static structure.

10 Conclusion

We have established a correspondence between software architectures and fiber bundles, enabling the application of characteristic class theory to architecture analysis. The first Chern class $c_1(E)$ measures coupling complexity; the first Stiefel-Whitney class $w_1(E)$ detects circular dependencies. The Whitney product formula enables compositional analysis: the complexity of a composed system is the product of its parts' complexities. Vanishing criteria confirm that acyclic (DAG) architectures are complexity-free.

The framework provides a principled vocabulary for architecture review that transforms qualitative judgments into computable, composable invariants. Characteristic classes can be tracked across codebase versions, enabling the detection of architectural drift. The accompanying Lean 4 formalization provides machine-checked proofs of all main results.

Within the Judgment Geometry series, this paper demonstrates that the sheaf-theoretic infrastructure developed in Papers 1–3 naturally extends to encode characteristic class invariants: the fiber bundle is a sheaf of local interface data, and the characteristic classes are global obstructions to trivializing this sheaf. The coupling complexity $c_1(E)$ is a cousin of the Čech cohomology obstructions studied in Paper 3; where Paper 3 obstructions live in $\check{H}^1$, coupling complexity lives in H^2.

Future work will develop:

- Automated tooling for computing MCI from Python codebases using AST analysis and type inference.
- A classification of architecture "diseases" by their characteristic class signatures.
- The full Grothendieck-Riemann-Roch theorem for software architectures, connecting maintenance cost to a topological index formula.

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